Lecture 21, Part III - Epidemiology

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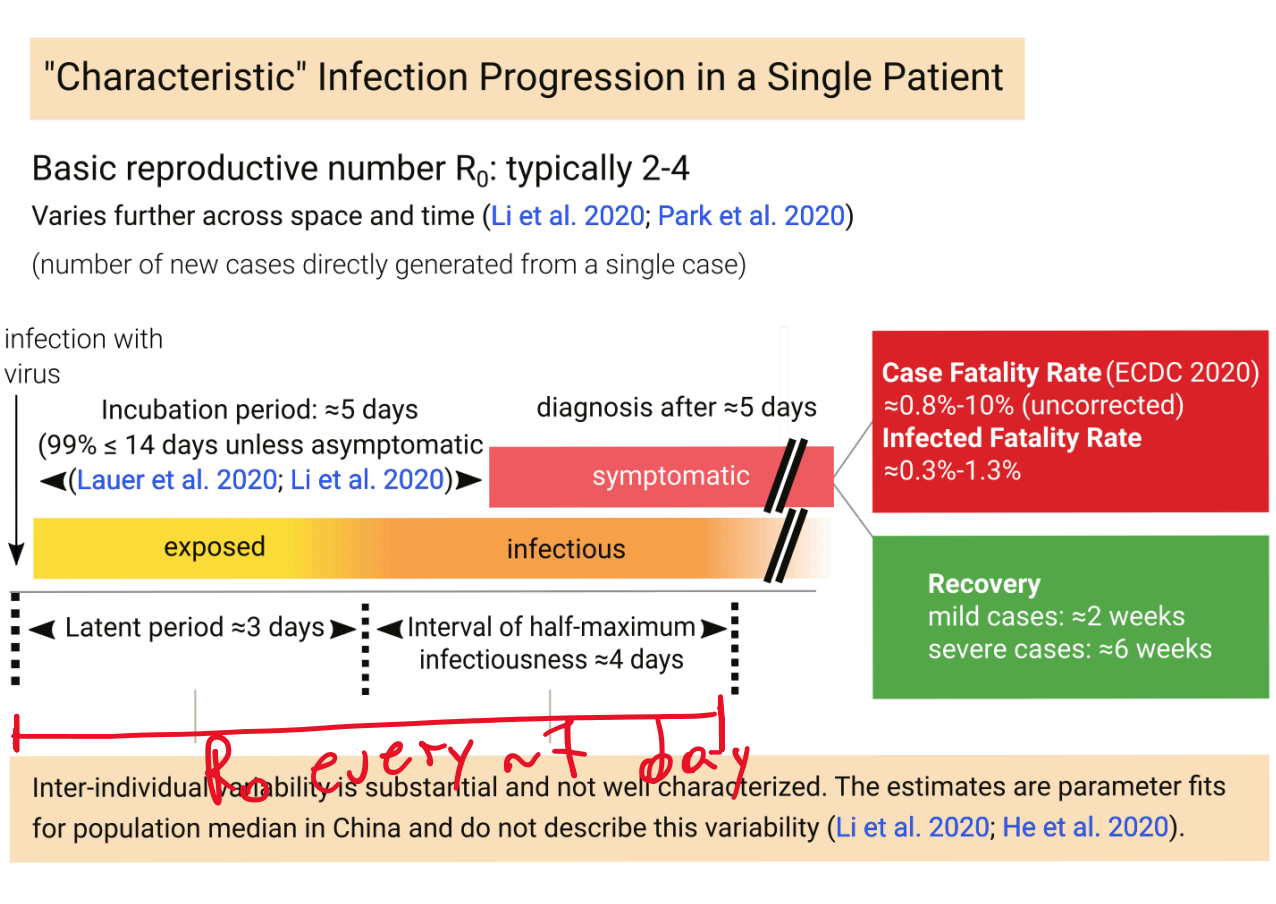
Models:
- ↳ Exponential
- ↳ Logistic: SI
- ↳ SIR → "flattening the curve"
- ↳ SIRQ → transients

Exponential growth:



"Characteristic" Infection Progression in a Single Patient

Basic reproductive number R_0: typically 2-4

Varies further across space and time (Li et al. 2020; Park et al. 2020)

(number of new cases directly generated from a single case)

Inter-individual variability is substantial and not well characterized. The estimates are parameter fits for population median in China and do not describe this variability (Li et al. 2020; He et al. 2020).

R_0 = basic reproductive number

= average # of people infected by one carrier in a susceptible population

$\simeq 2 \sim 4$

We have R_0 every 7 days

$\tau = 7$ days

Guided struggle: Calculate # of infected vs. t as a function of R_0 and τ assuming $N_{inf}(t=0)=1$. Plot $N_{inf}(t)$ for $R_0 = 2$ and 4. How long to get to US population? world pop?

$$N_{infected} = R_0^{t/7\,days}$$

(if we start from one individual)

$$\ln(N_{inf}) = \frac{t}{\tau}\ln R_0$$

$$\tau\frac{\ln(N_{inf})}{\ln(R_0)} = t$$

Logistic growth:

SI model:

S I

$S + I = N$

"mass action"

$$\text{prob. of an interaction} = \overset{\text{either draw S or I first}}{2}\underbrace{\frac{S}{N}}_{\text{prob of susc.}}\cdot\underbrace{\frac{I}{N-1}}_{\text{prob. of infected}} \approx 2\frac{S}{N}\cdot\frac{I}{N}$$

$\Rightarrow$ rate of infection $\propto S \cdot I$

$$I(t+\Delta t) = I(t) + r\cdot S\cdot I\,\Delta t$$

$$\frac{dI}{dt} = r\cdot SI = r\,I\,(N-I)$$

$\downarrow$

$N = S + I$

$S = N - I$

$$\frac{dI}{dt} = \underbrace{rN}_{\text{early growth rate}} I\left(1 - \frac{I}{\boxed{N}}\right)$$

N ↓ carrying capacity

Logistic growth!

SIR model:

Guided struggle: write the corresponding system of equations

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

If I figure out these dynamics

$$\frac{dR}{dt} = \gamma I$$

I can calculate R

$\downarrow$

$$\text{removal rate} = \frac{1}{\text{time for recovery}} \simeq \frac{1}{14\text{ days}}$$

So we need to solve the dynamics of S and I.

To make progress, let's make the system non-dimensional:

① Divide the eqns. by γ

$$\frac{dS}{dt}\frac{1}{\gamma} = -\frac{\beta}{\gamma}SI$$

$$\frac{dI}{dt}\frac{1}{\gamma} = \frac{\beta}{\gamma}SI - I$$

② Measure time in units of $1/\gamma \Rightarrow \tau = t\gamma$

Define the ratio

$$\frac{\beta}{\gamma} = r$$

③ The new system is

$$\frac{dS}{d\tau} = -rSI$$

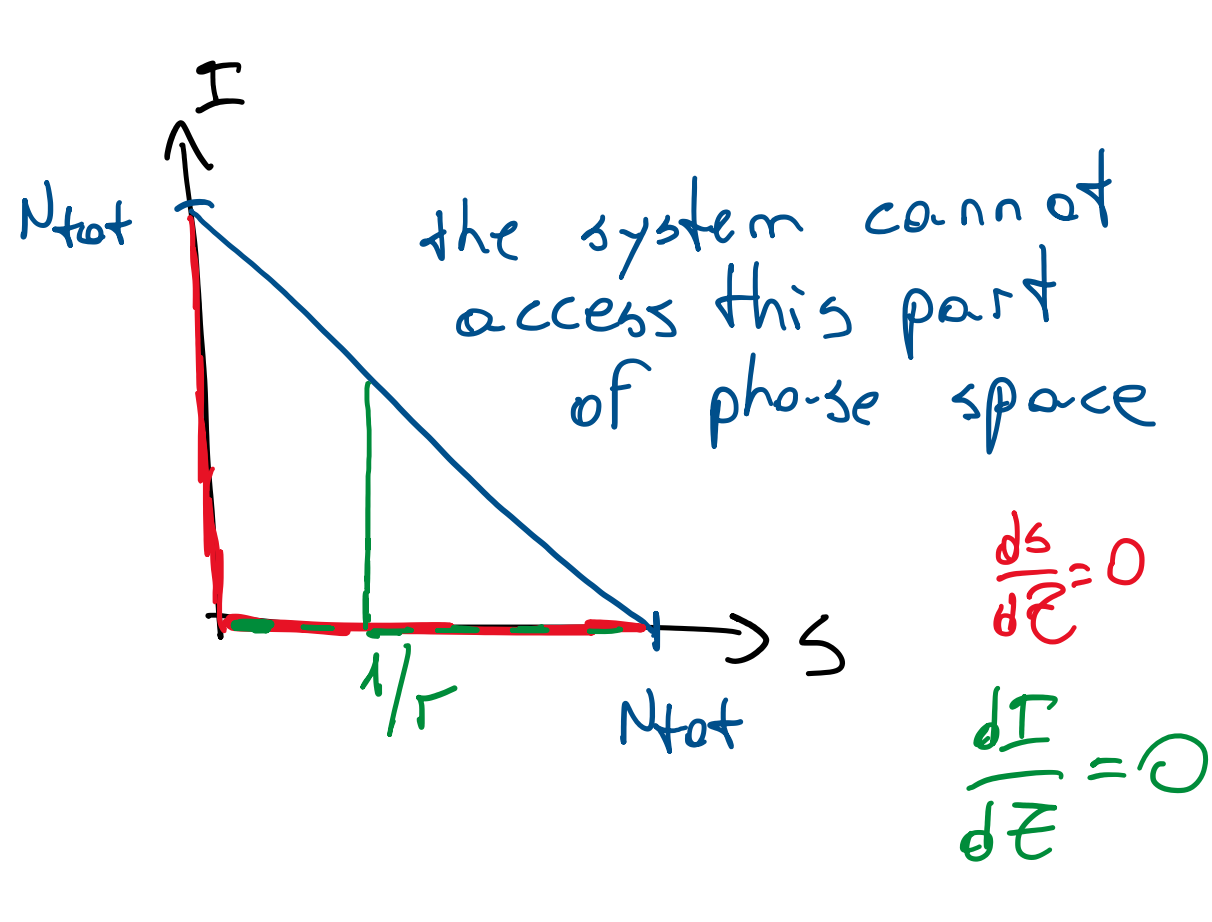
$$\frac{dI}{d\tau} = rSI - I$$

Note that now I have only one parameter dictating the dynamics

What can we learn from the phase diagram?

↳ Remember that $S + I + R = N_{tot}$

$\Rightarrow S + I < N_{tot}$



Find the nullclines:

$$\frac{dS}{d\tau} = -rSI = 0 \Rightarrow \boxed{S=0} \text{ or } \boxed{I=0}$$

$$\frac{dI}{d\tau} = rSI - I = 0$$

$$rSI = I \Rightarrow \boxed{S = \frac{1}{r}}$$

or

$$\boxed{I = 0}$$

Note that at any point, if $I = 0$ then $\frac{dS}{d\tau}$ and $\frac{dI}{d\tau} = 0$. This should be a stable equilibrium point. No infected = new infection!

The question is: what are the dynamics of the system to get to this point?

Let's plot the vector field:

$$\frac{dS}{d\tau} = -rSI \Rightarrow \frac{dS}{d\tau} < 0 \text{ for all } S \text{ and } I.$$

$$\frac{dI}{d\tau} = rSI - I = I(S - \tfrac{1}{r})$$

$\Rightarrow$ if $rS - 1 < 0$

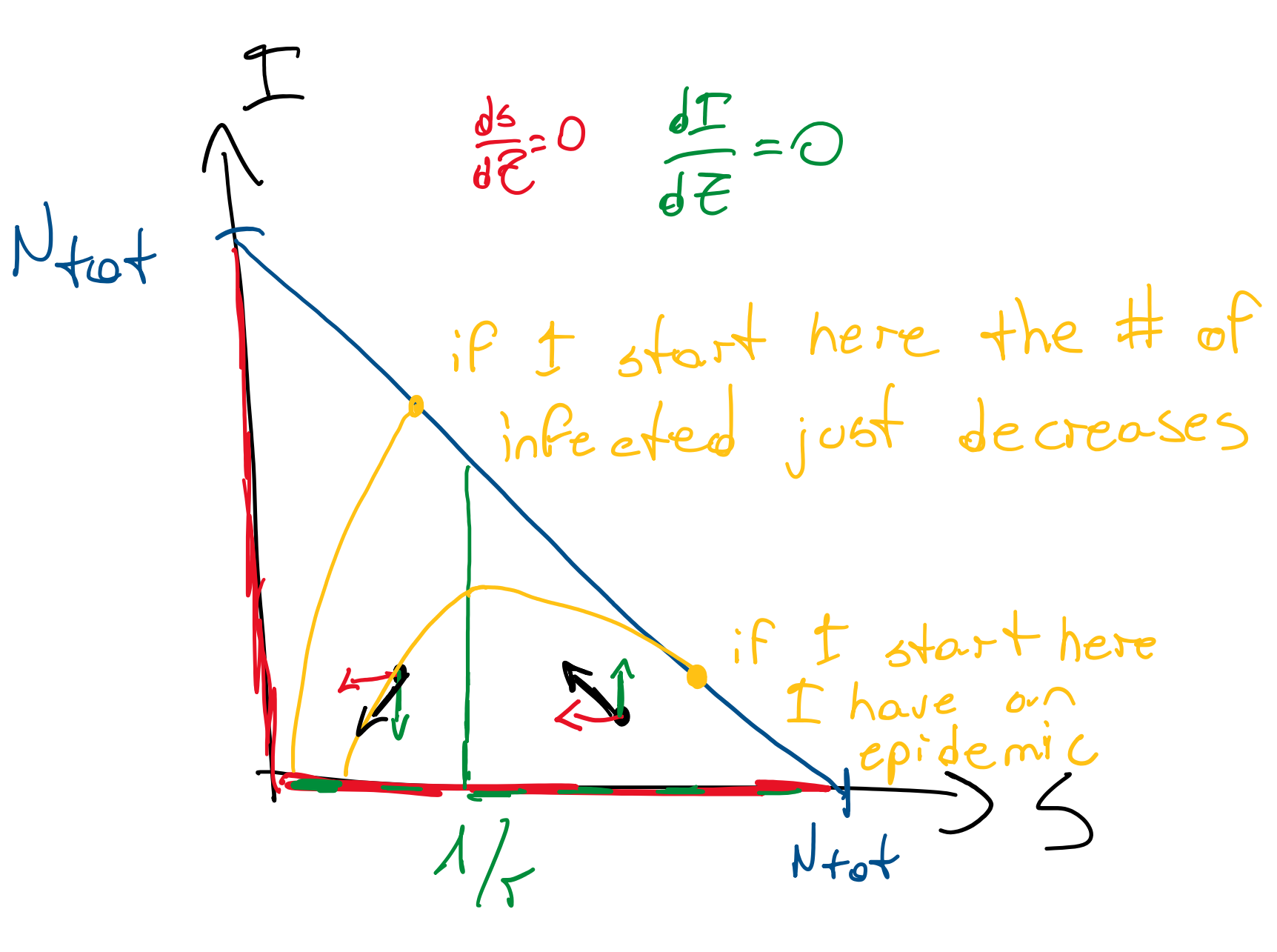
$S < 1/r$

then $\frac{dI}{d\tau} < 0$

if $rS - 1 > 0$

$S > 1/r$

then $\frac{dI}{d\tau} > 0$



I
dS/dτ = 0
dI/dτ = 0
Ntot
if I start here the # of infected just decreases
if I start here I have an epidemic
1/r
Ntot
S

To put this differently, let's examine the behavior early on such that at $t=0$: $S(t=0) = S_0$

$$\left.\frac{dI}{d\tau}\right|_{t=0} = I\,\underbrace{(rS_0 - 1)}$$

>0, I'll have an increase in I = epidemic

<0, I will decrease

$\Rightarrow$ No epidemic: $rS_0 - 1 < 0$

$$\underbrace{rS_0}_{R_0} < 1$$

R_0 = basic reproduction number

Guided struggle: Plot the vector field.

Let's use $R_0 = 2$ with $N_{tot} = 8E6$, $S_0 = 8E6$

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